COMPUTER NETWORKS

CSE 321

HTTP Continued

Request Methods

Specifying the purpose of the request

Request Methods

GET

Retrieves/Reads data from a web server. Expected to contain ‘no body’.

Request Methods

POST

Sends data to the server. Normally carries data in the payload.

Request Methods

PUT

Modifies the data on the server.

Request Methods

DELETE

Removes a resource on the server.

Request Methods

HEAD

Retrieves the header as GET request would retrieve without the actual data. Helps to check the existence and metadata of the resources.

And many more...

Response Status Code

Shows whether the request was successful and helps to take necessary steps.

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200 - OK

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Response Status Code

Shows whether the request was successful and helps to take necessary steps.

200 - OK

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401 - Unauthorized

404 - Not Found

Response Status Code

Shows whether the request was successful and helps to take necessary steps.

200 - OK

201 - Created

400 - Bad Request

401 - Unauthorized

404 - Not Found

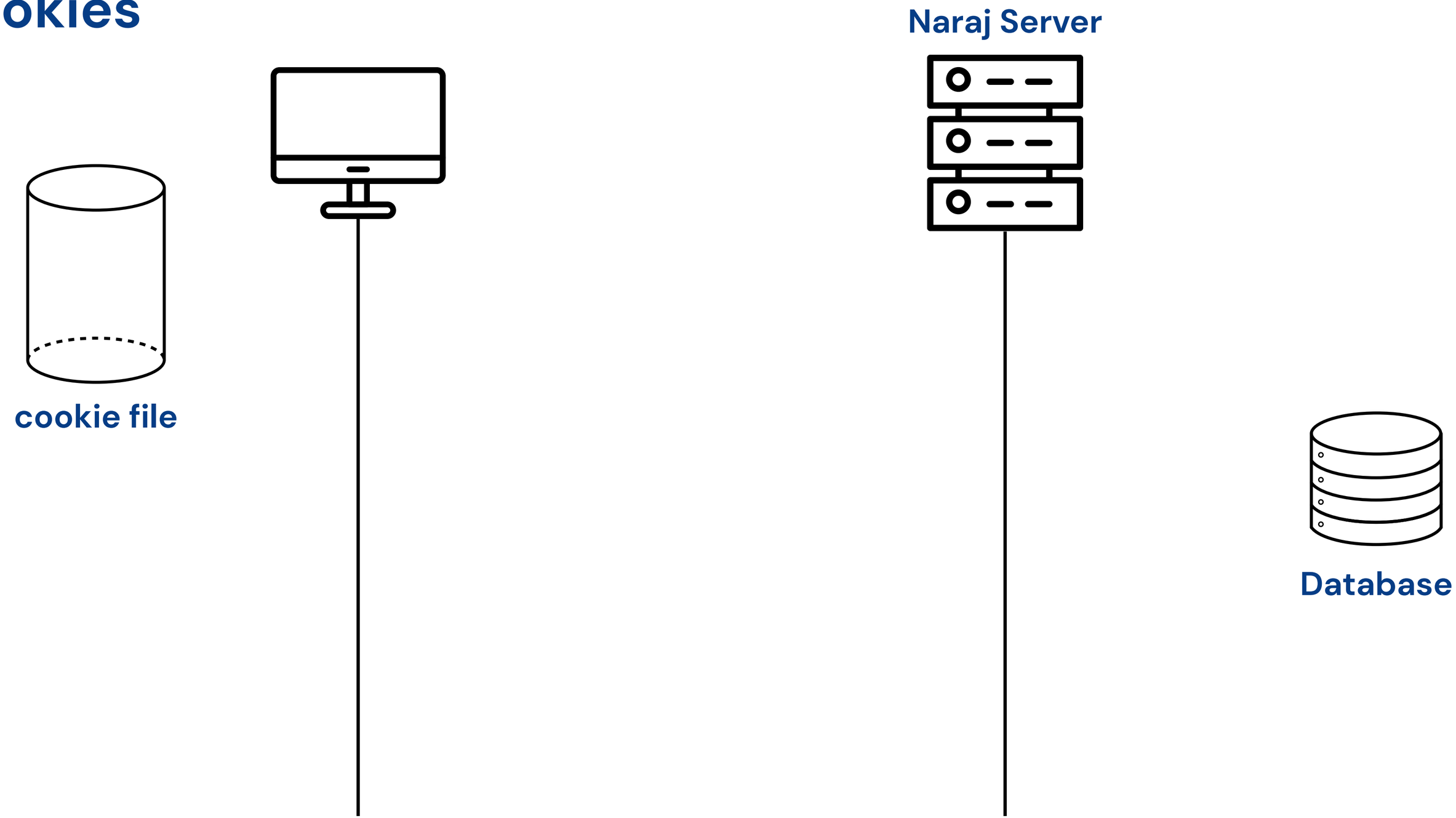
500 - Internal Server Error

...

Maintaining States

HTTP is a stateless protocol. So how it is remembering its clients? (e.g. when we try to visit the same website after a short time of logging in, we are able to get in without log-in.

Cookies



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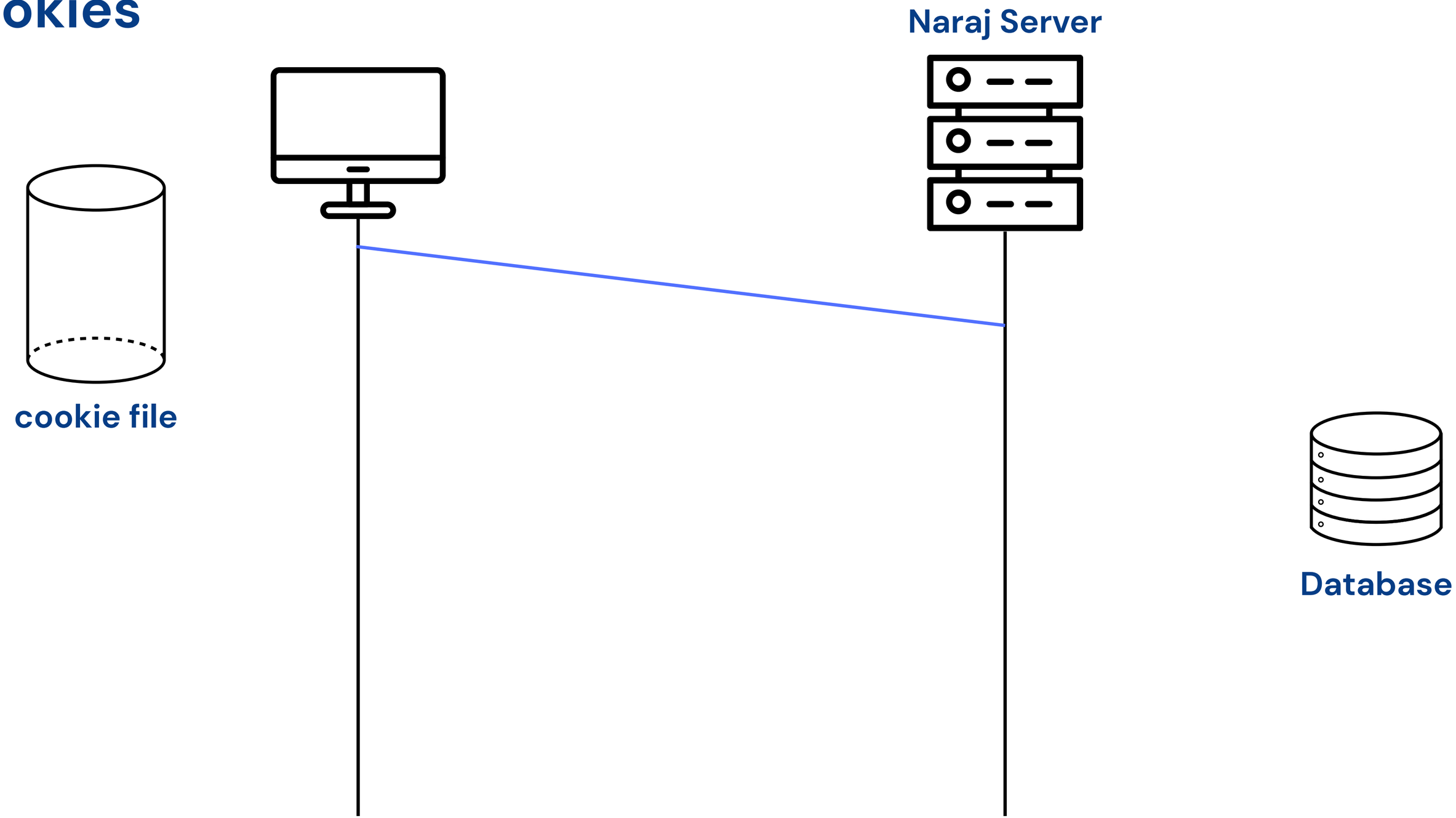

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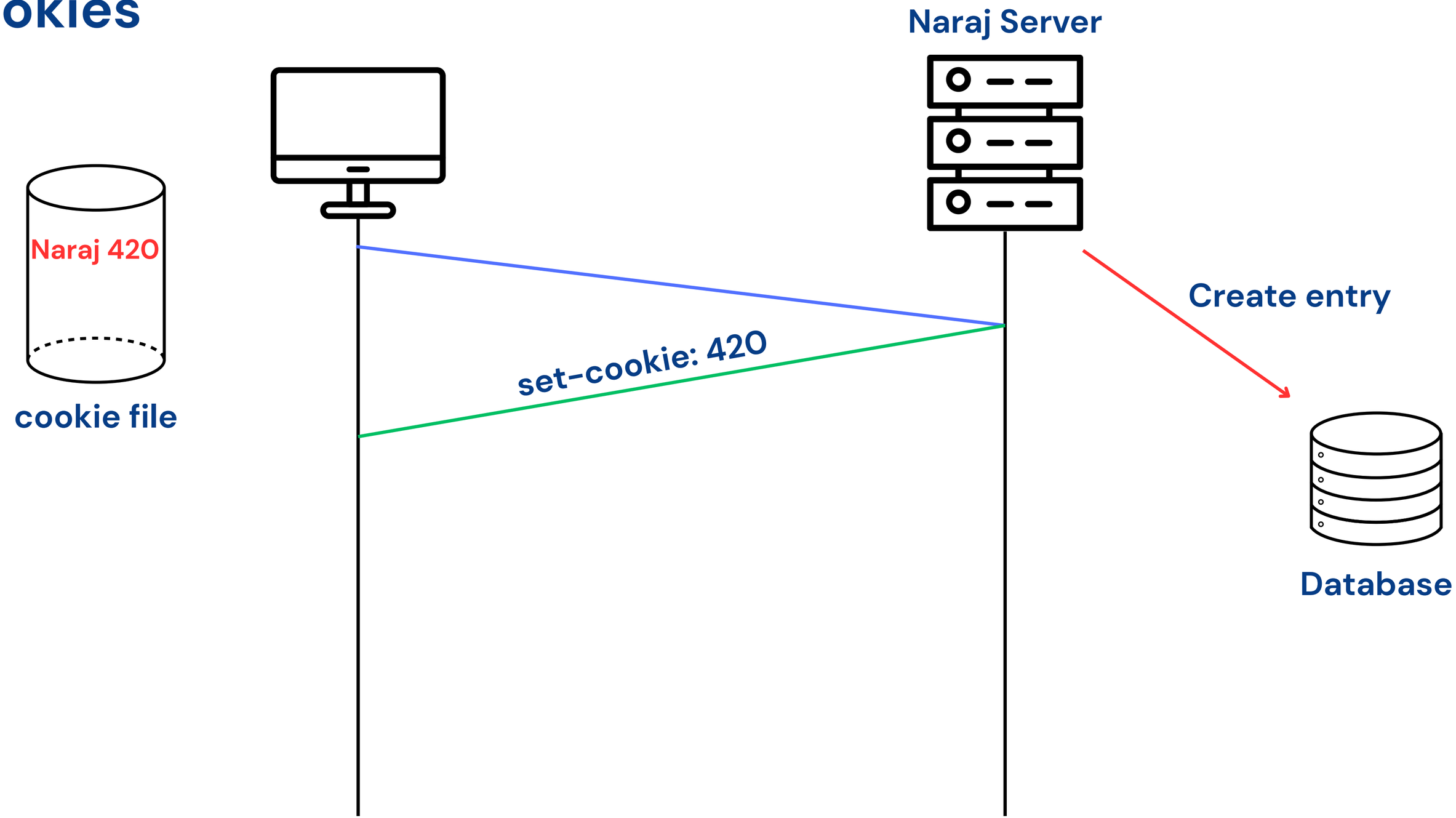
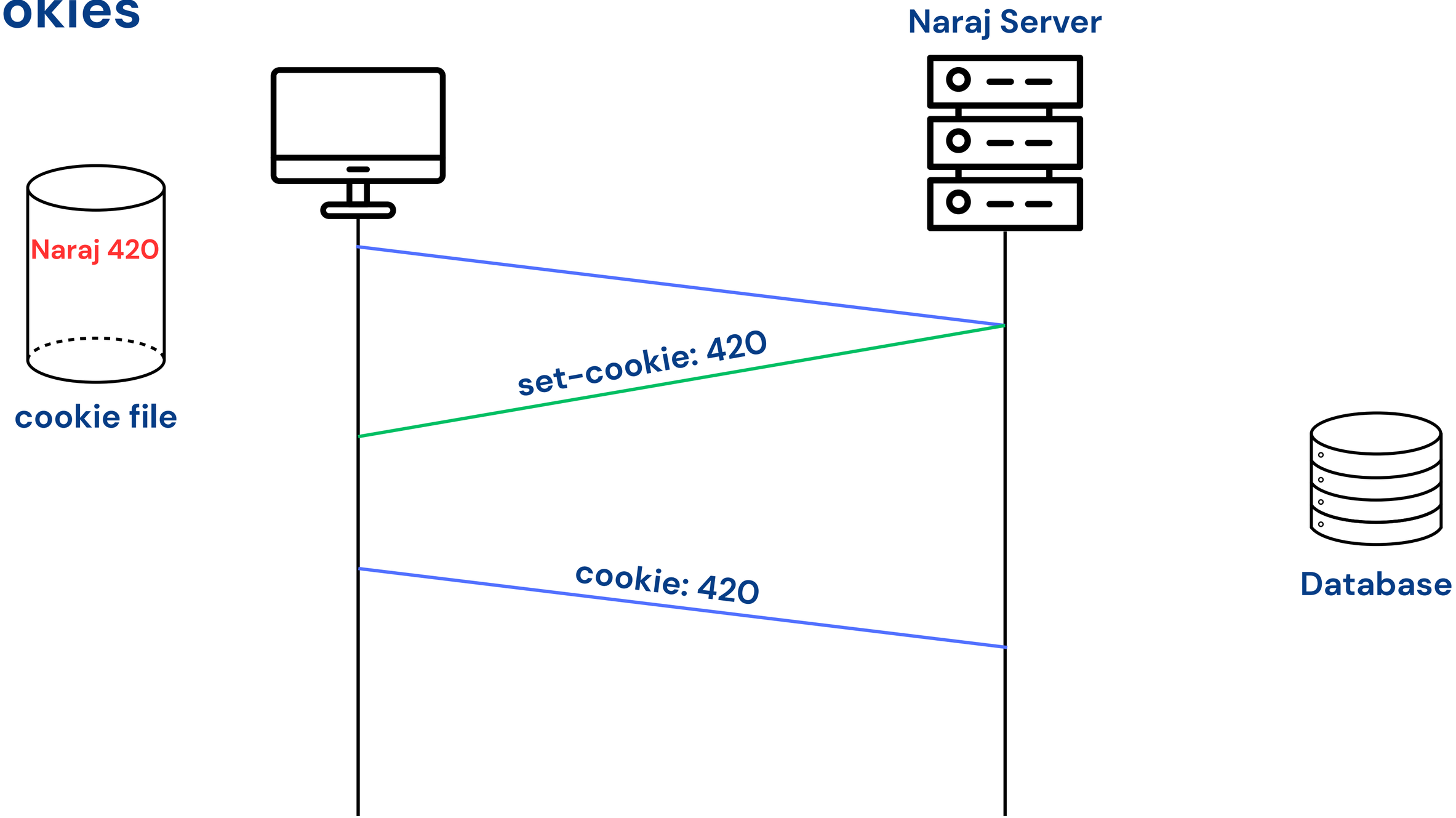

Naraj Server
Naraj 420
cookie file
set-cookie: 420
Create entry
Database

Cookies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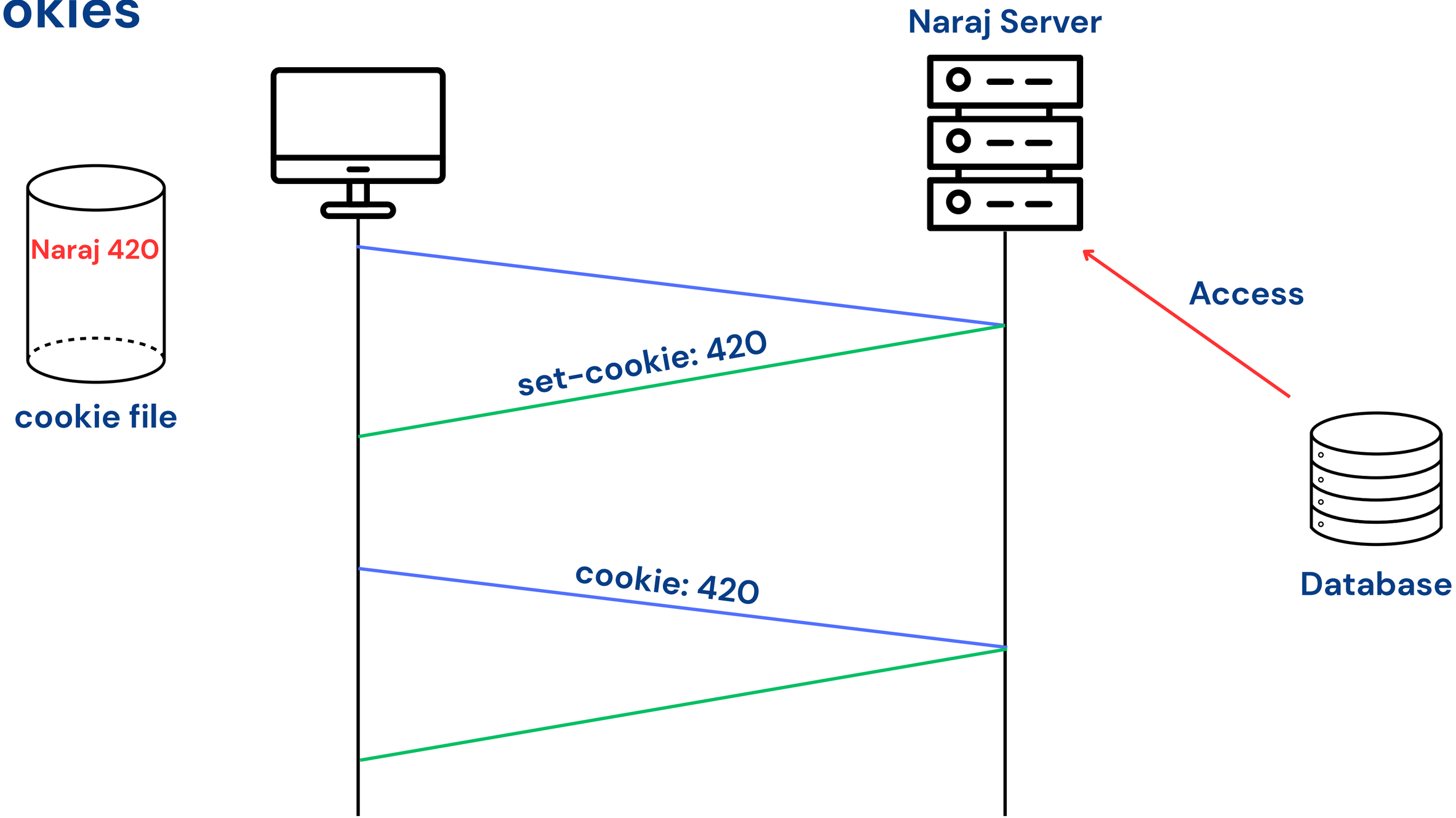

Naraj Server
Naraj 420
cookie file
set-cookie: 420
cookie: 420
Access
Database

Cookies

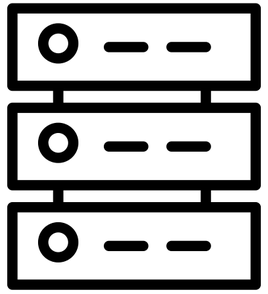
Can be used to store:

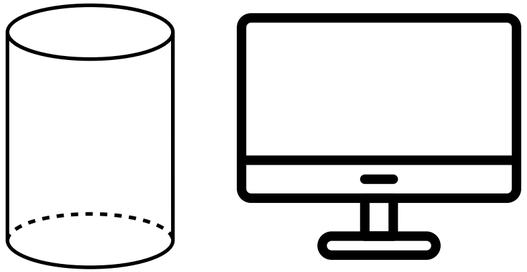
- Session ID
- Authentication token
- User preferences (like language, theme etc.)
- Shopping cart info

etc...

How Cookies Track Users

prothom kalo





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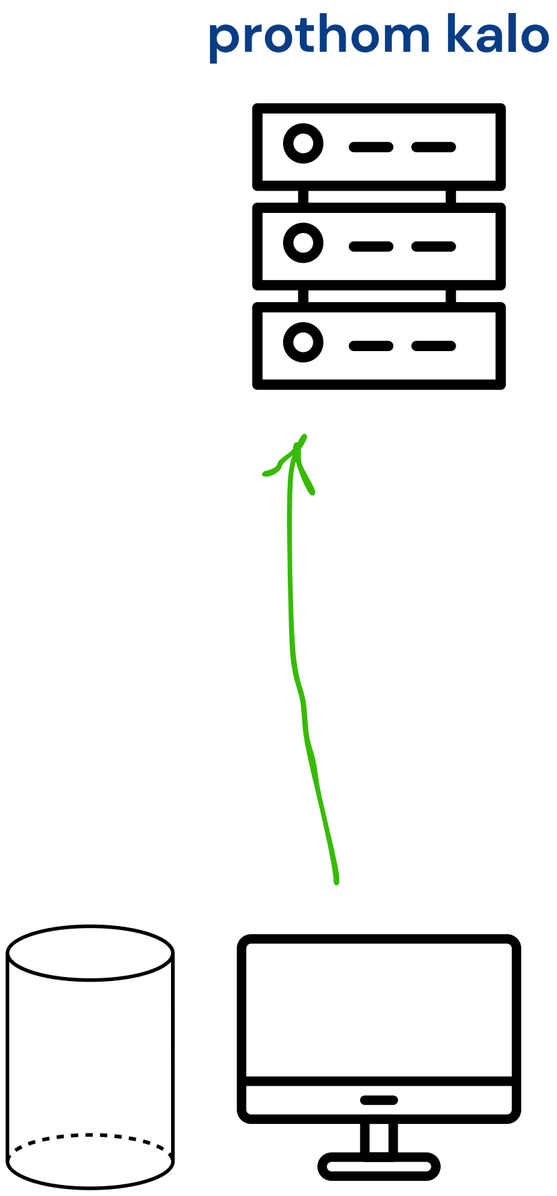

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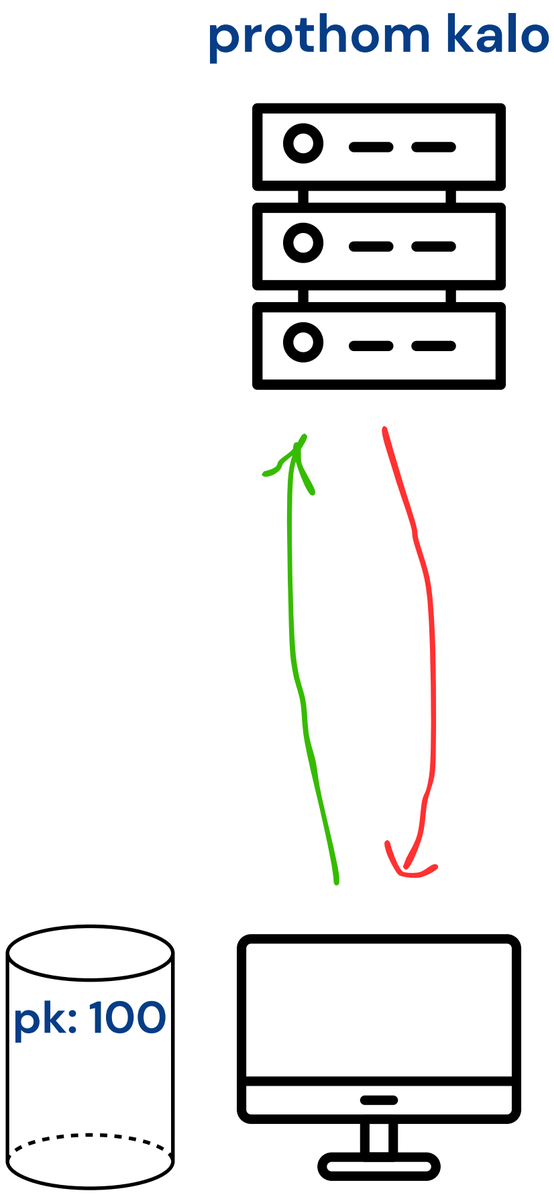

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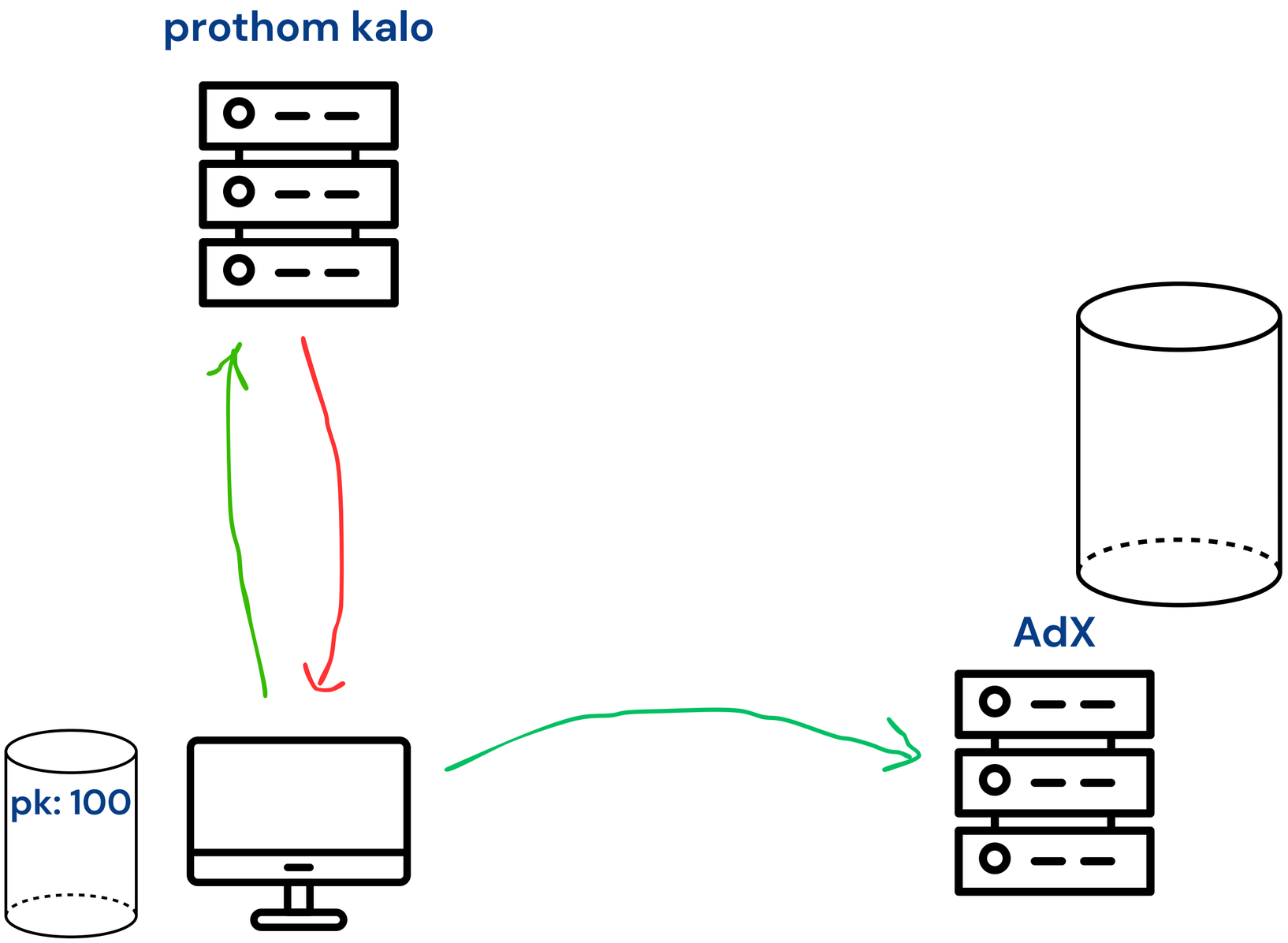

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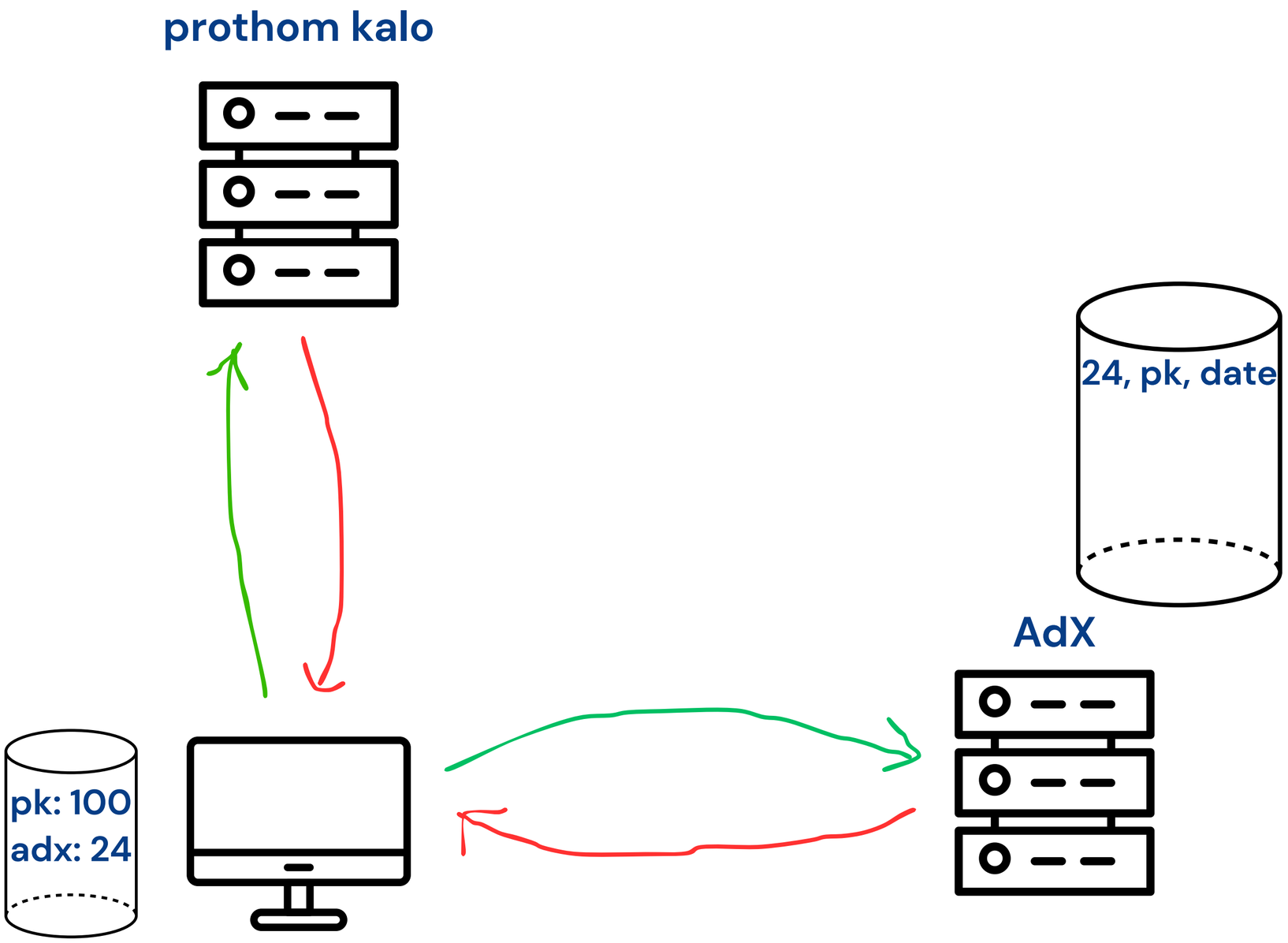

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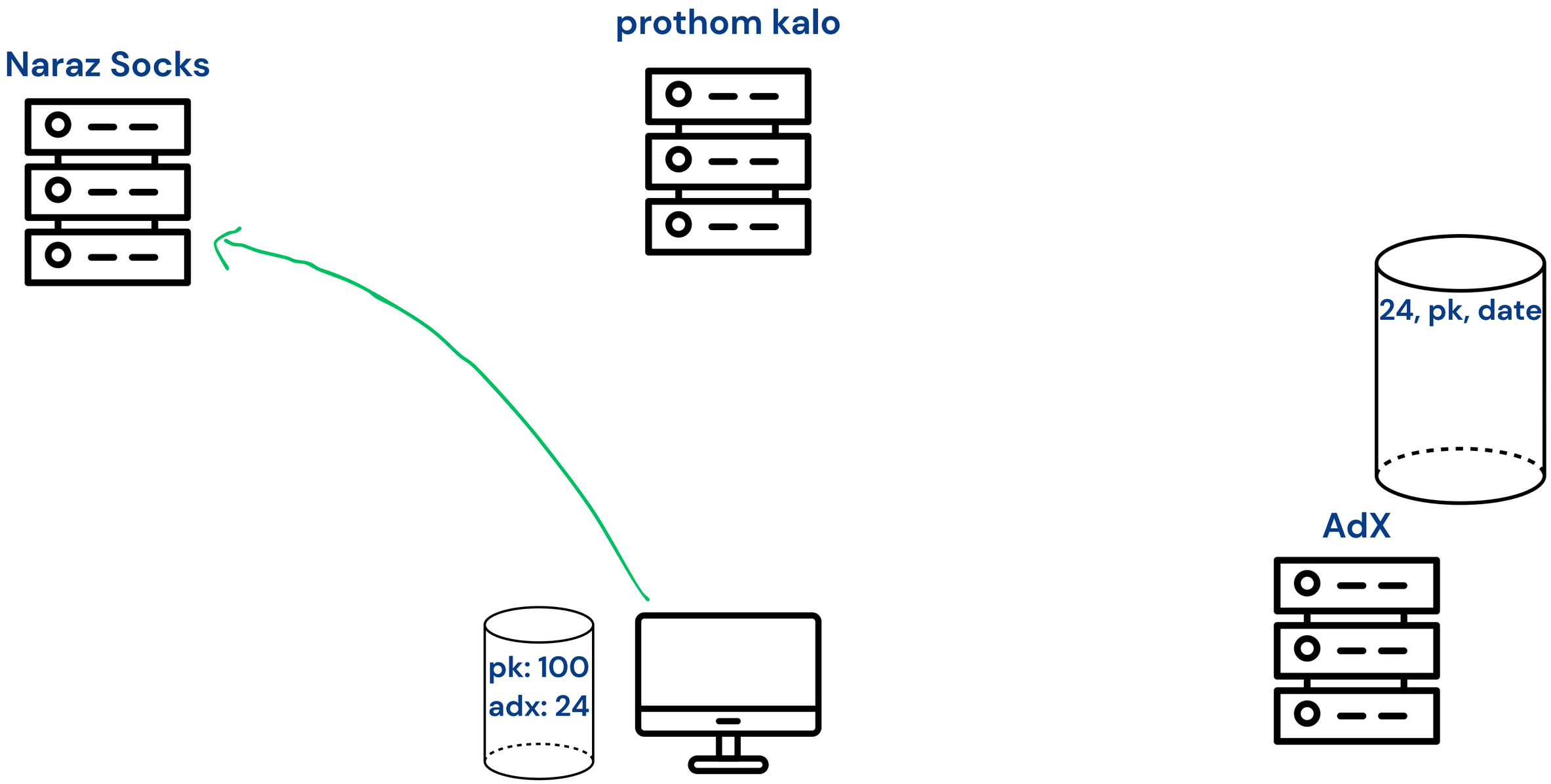

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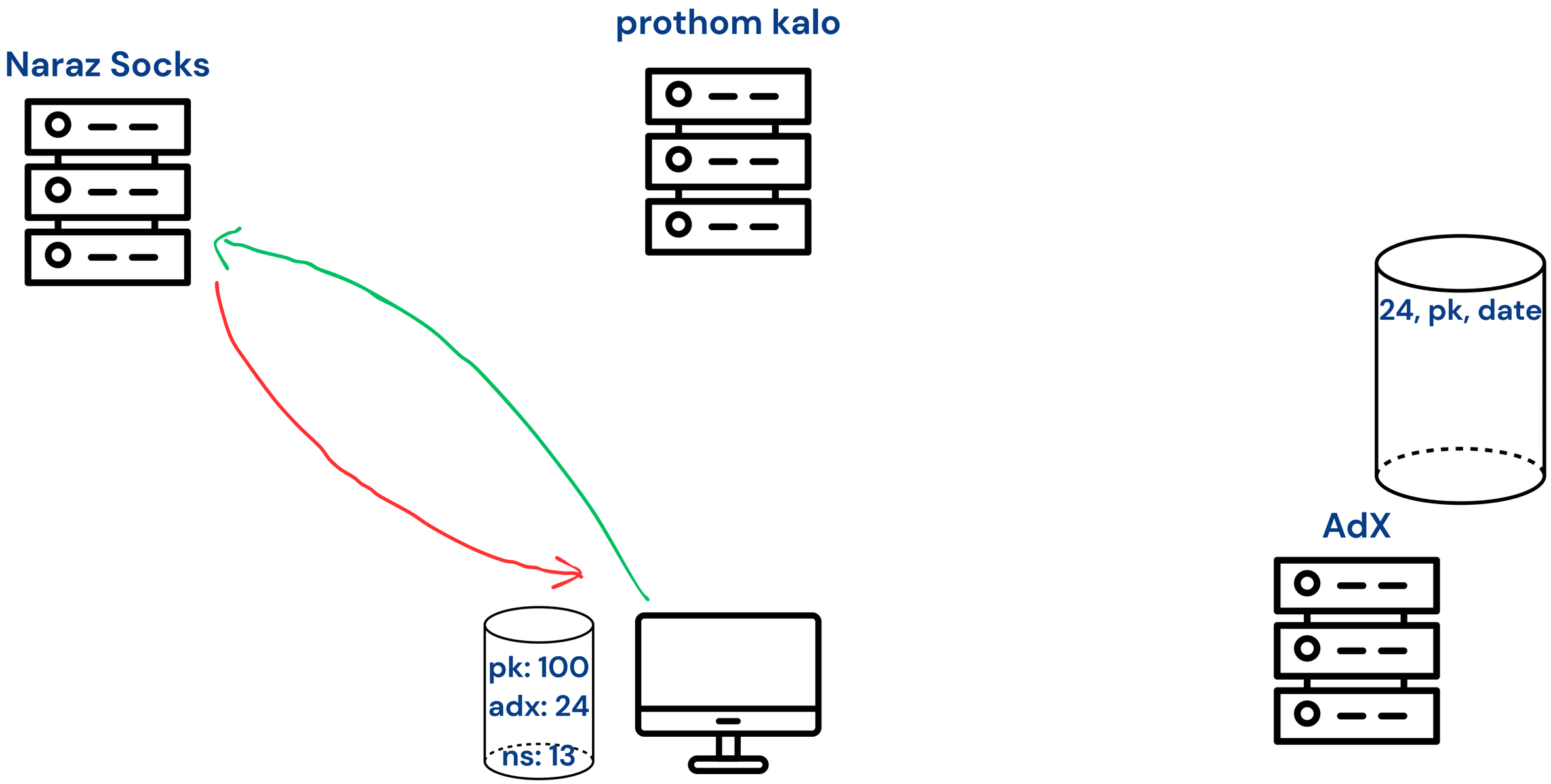

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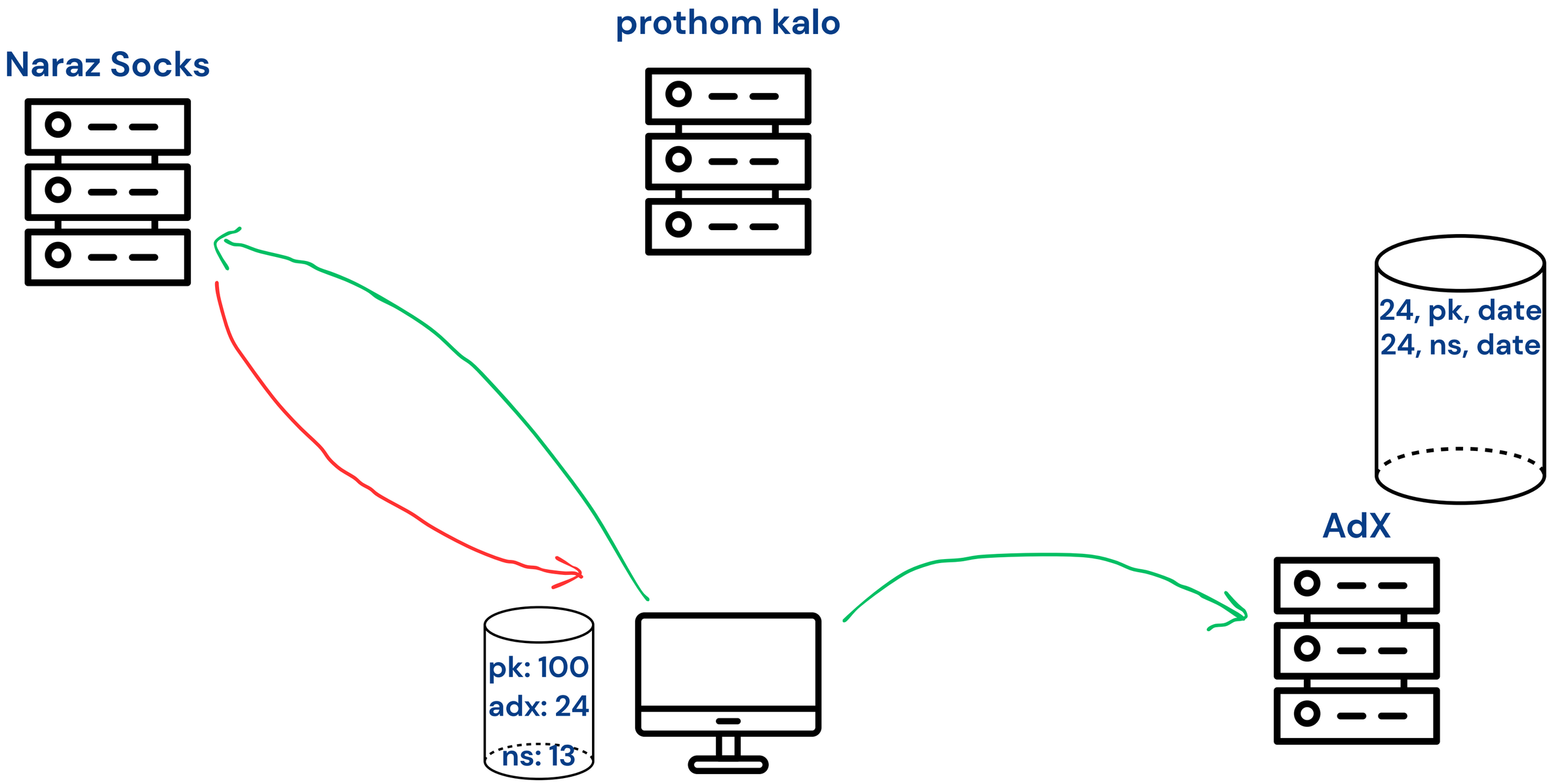

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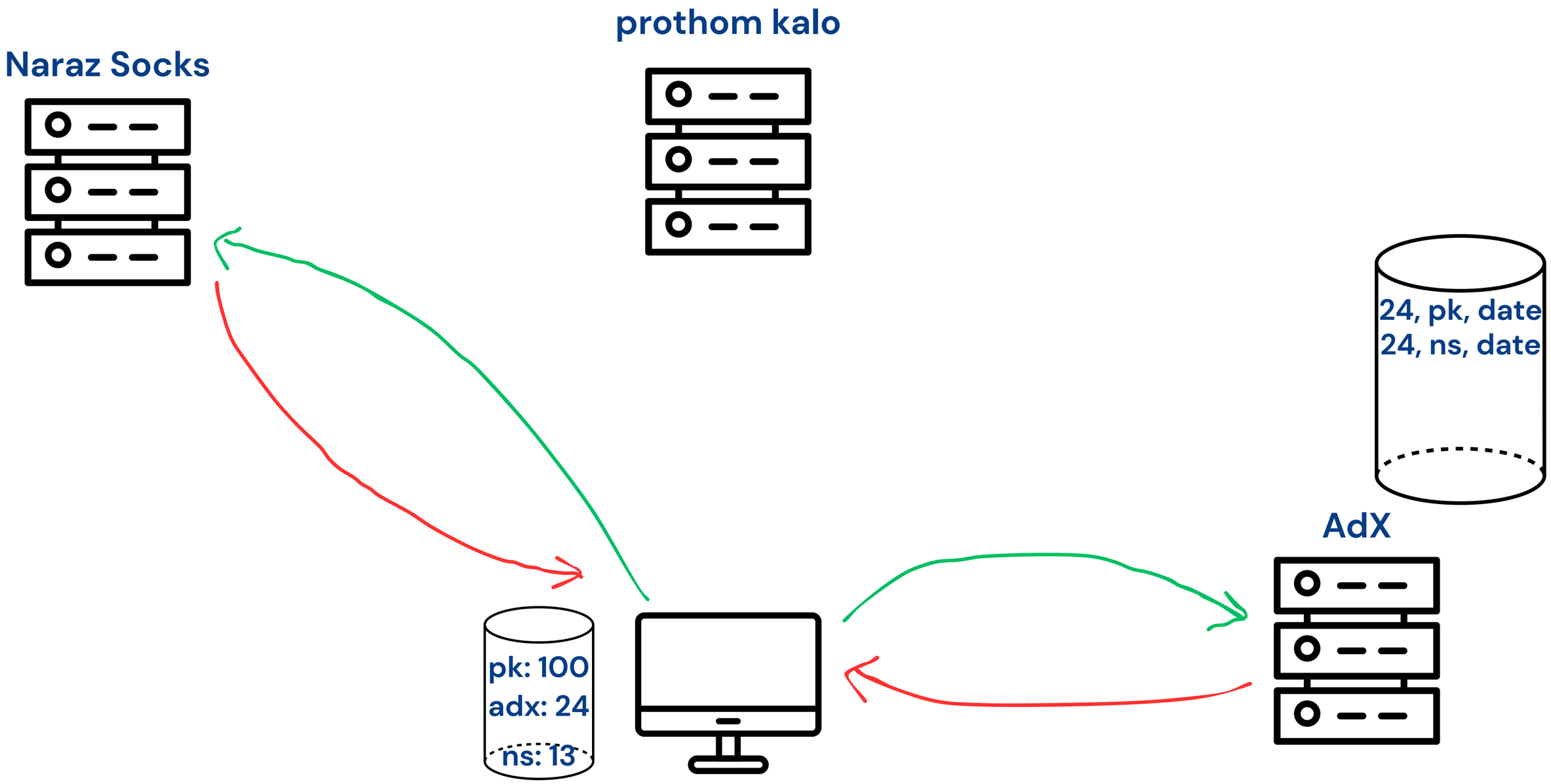
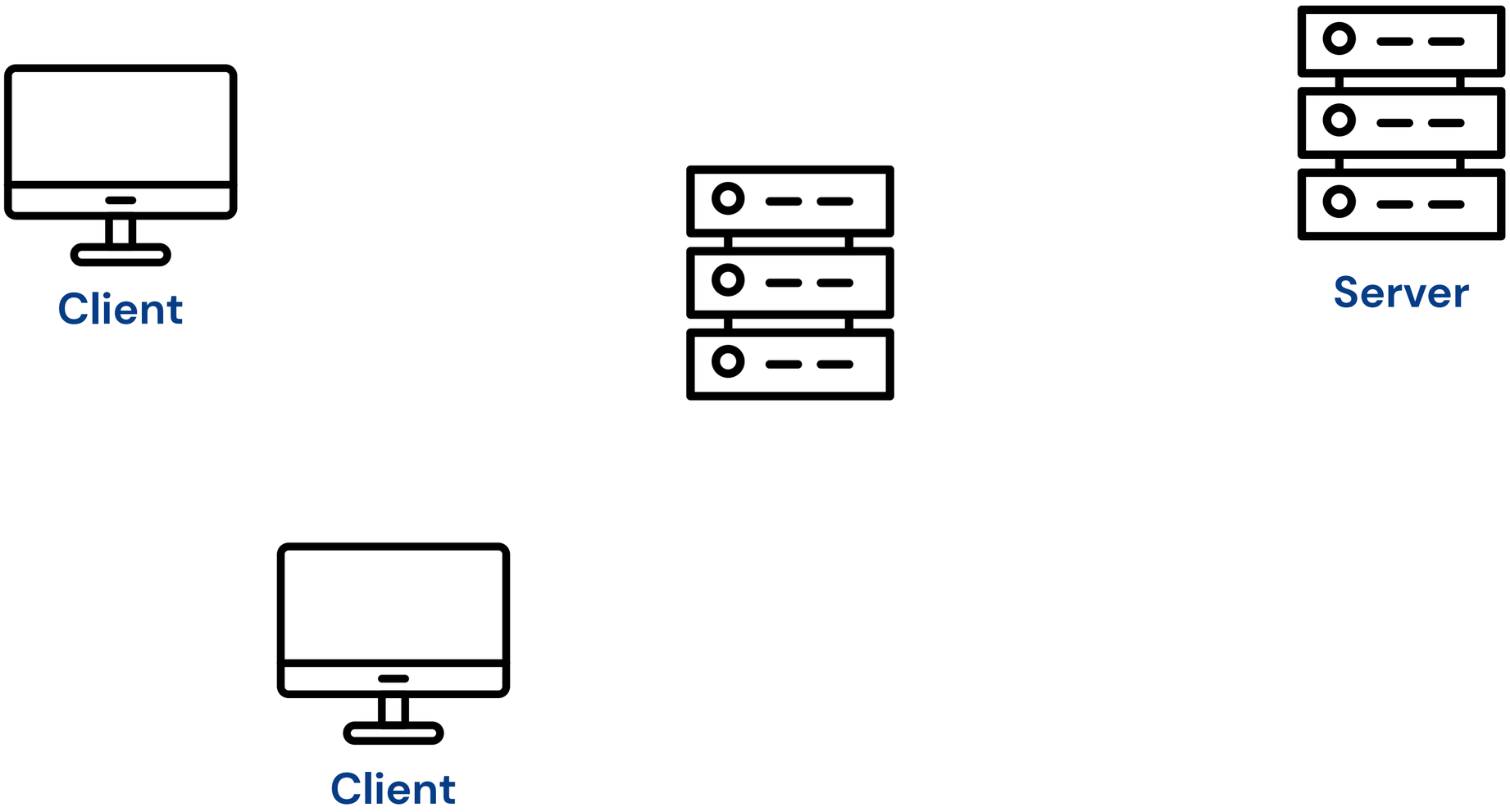


Increasing Performance (Web Cache)



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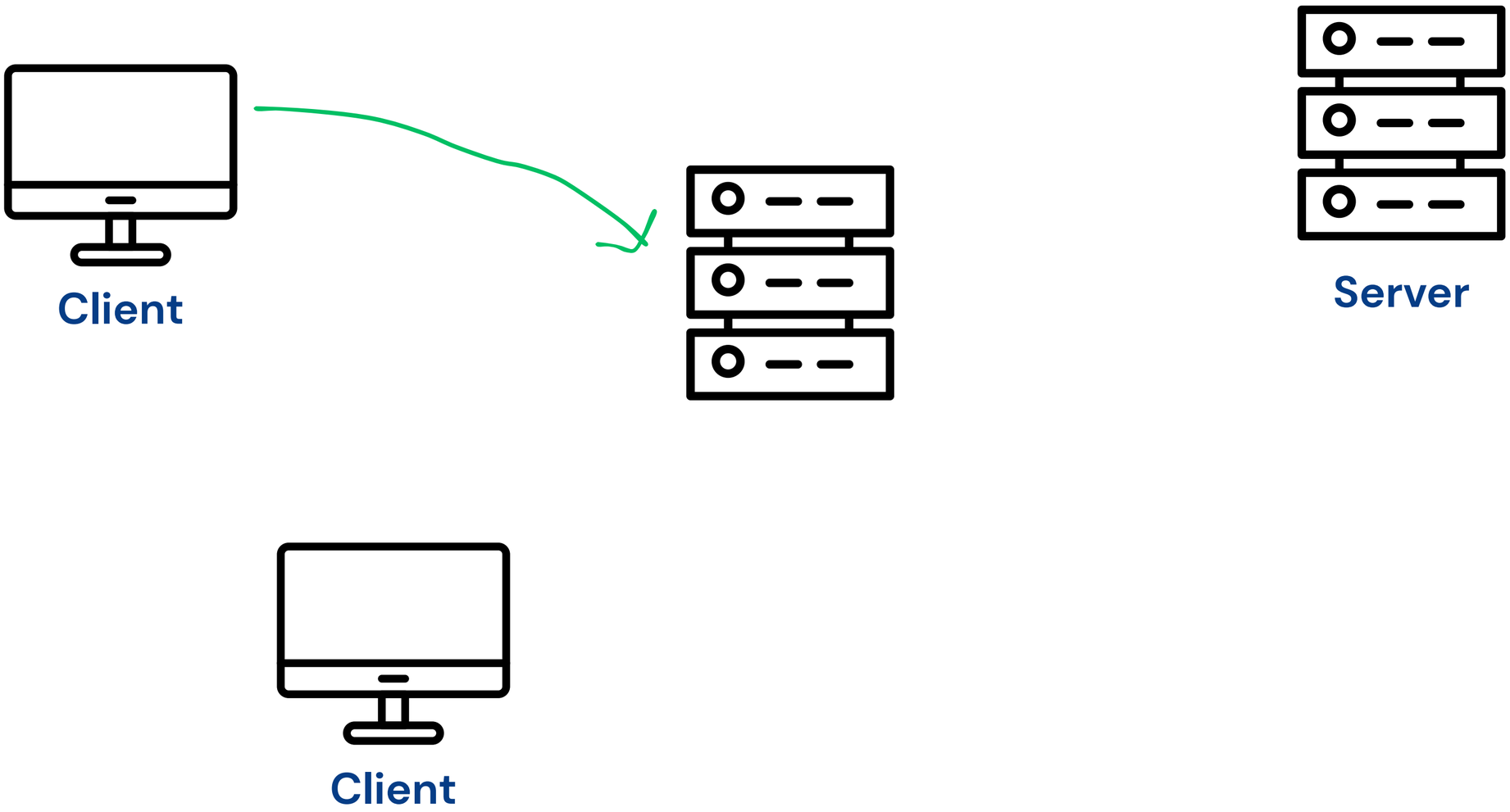

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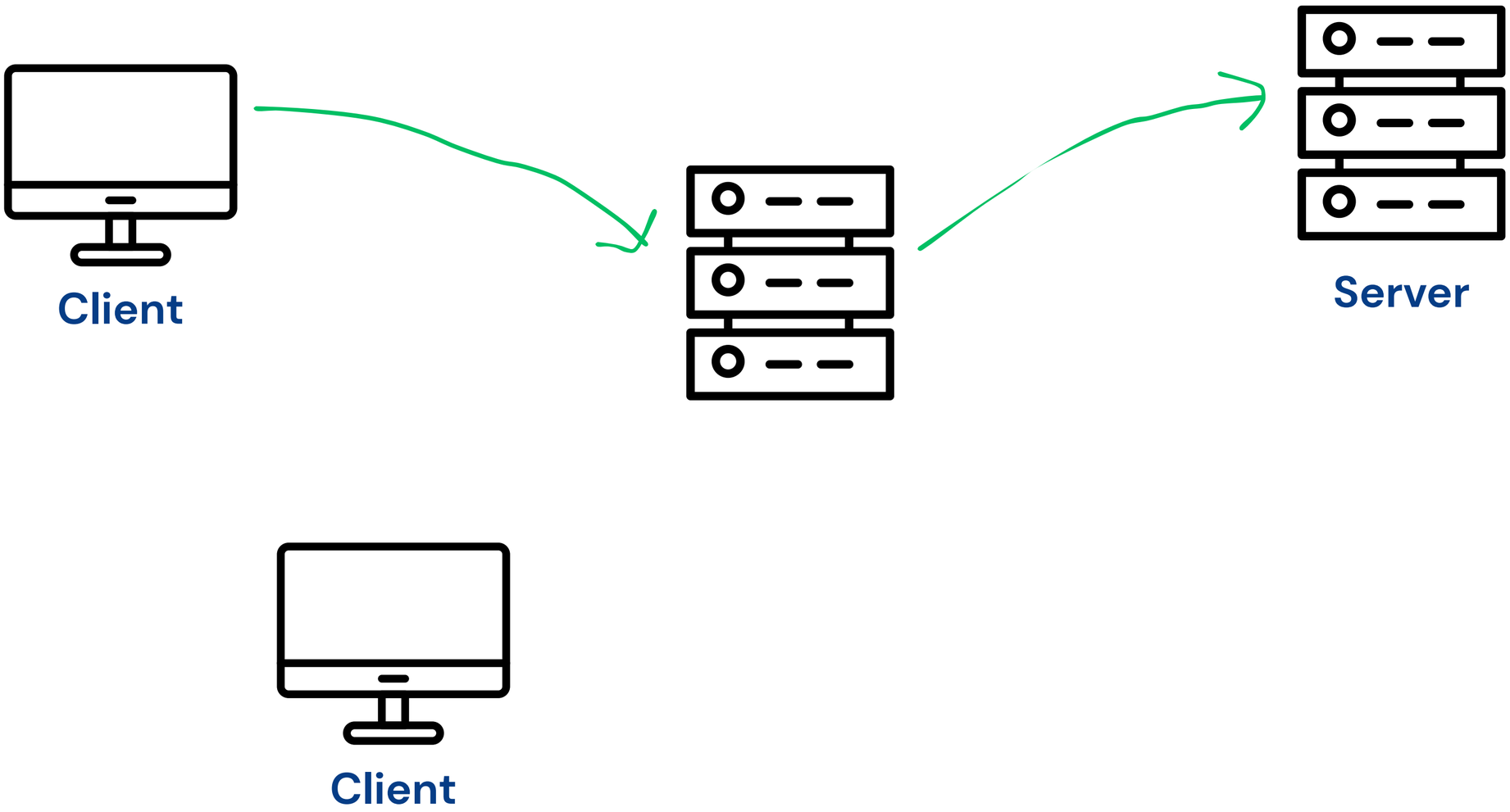

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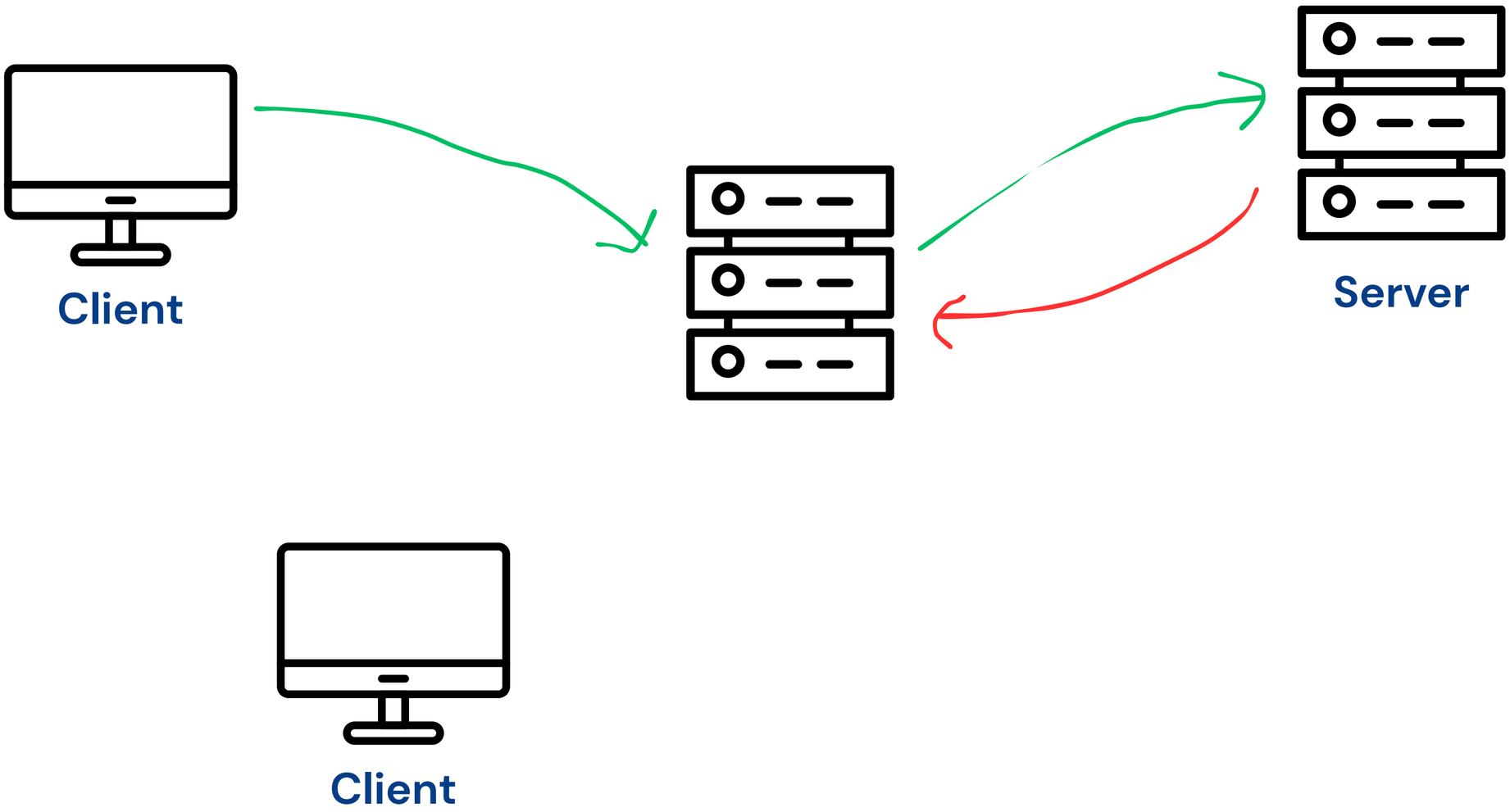

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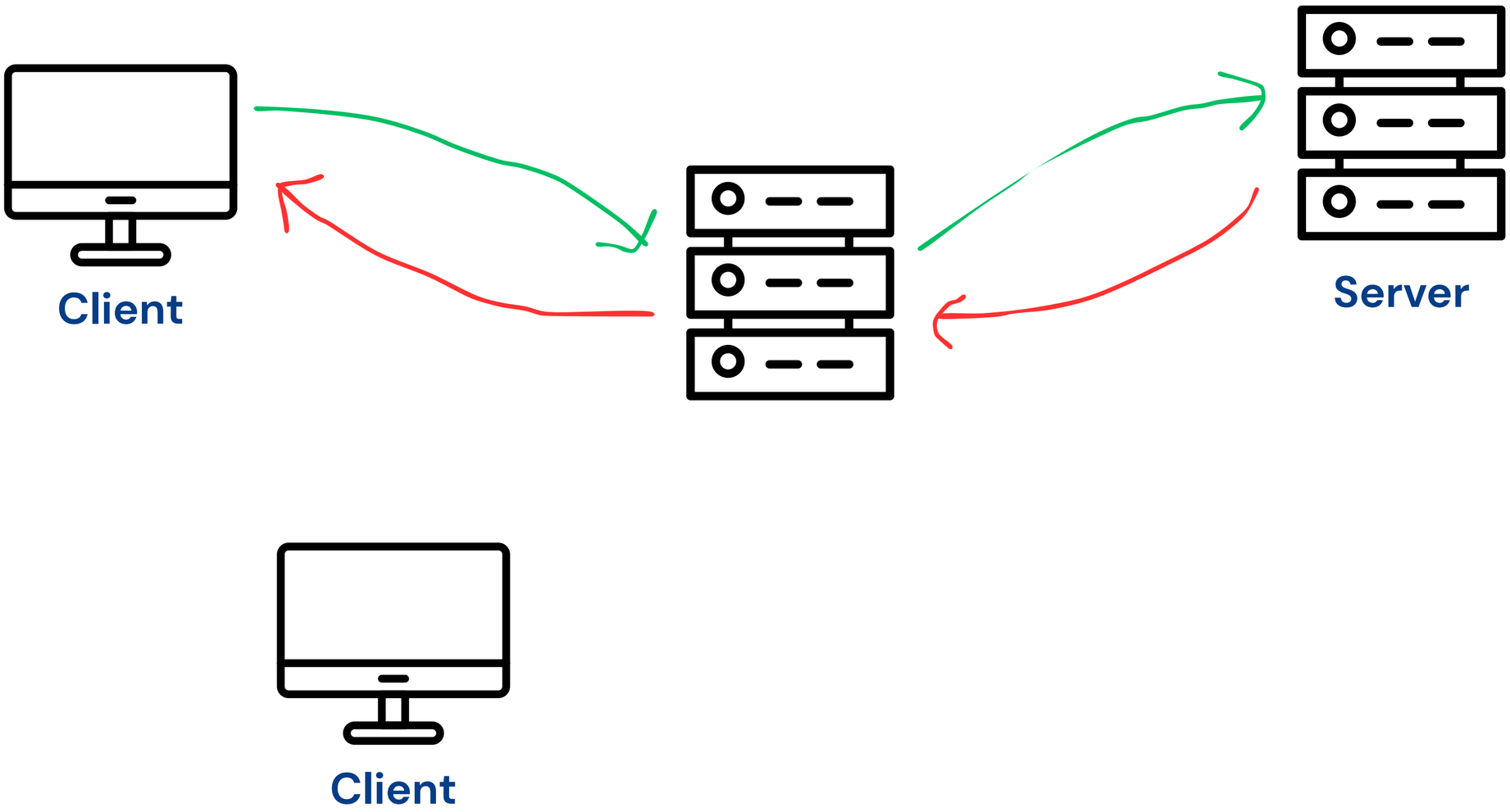

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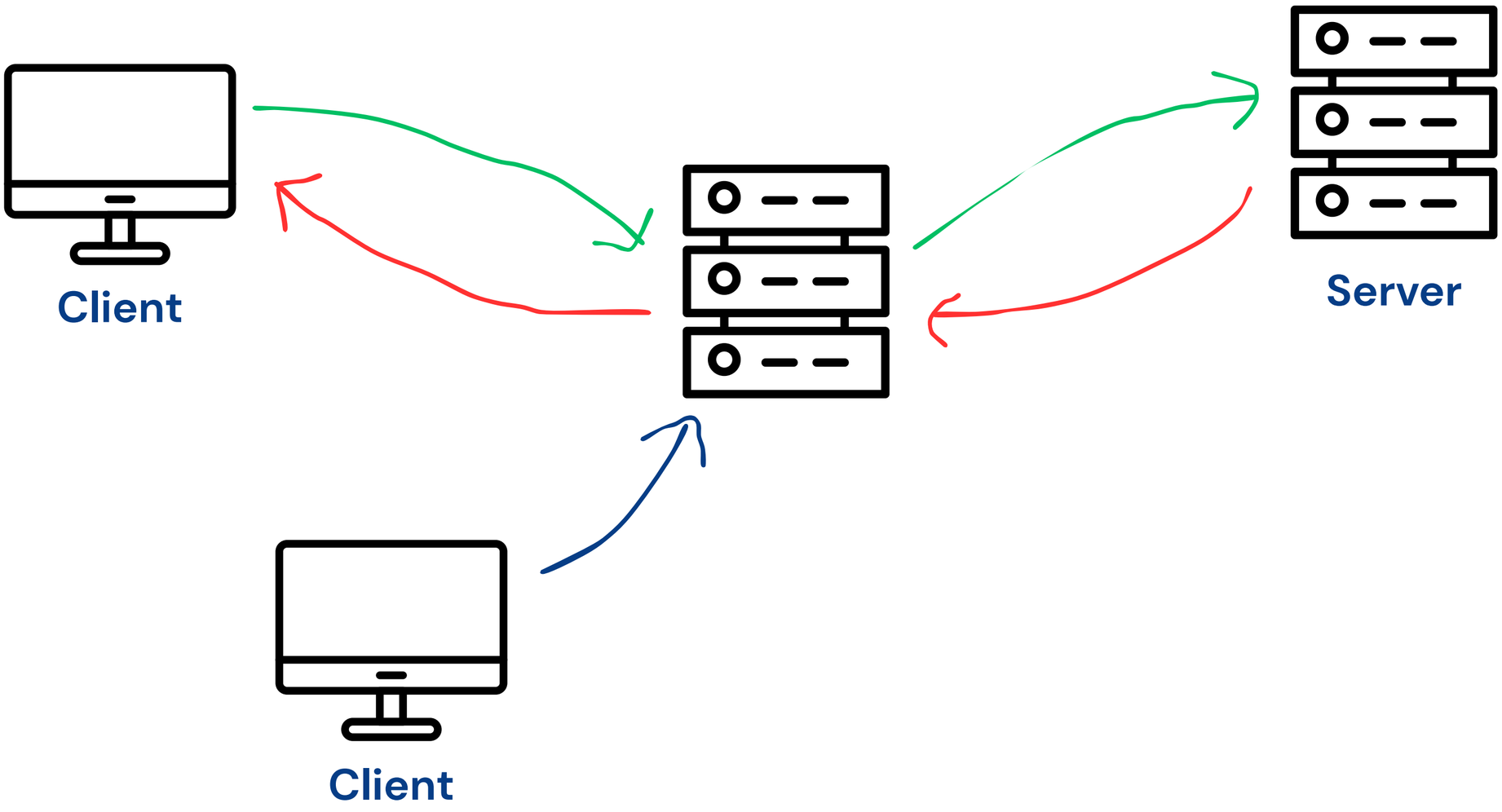

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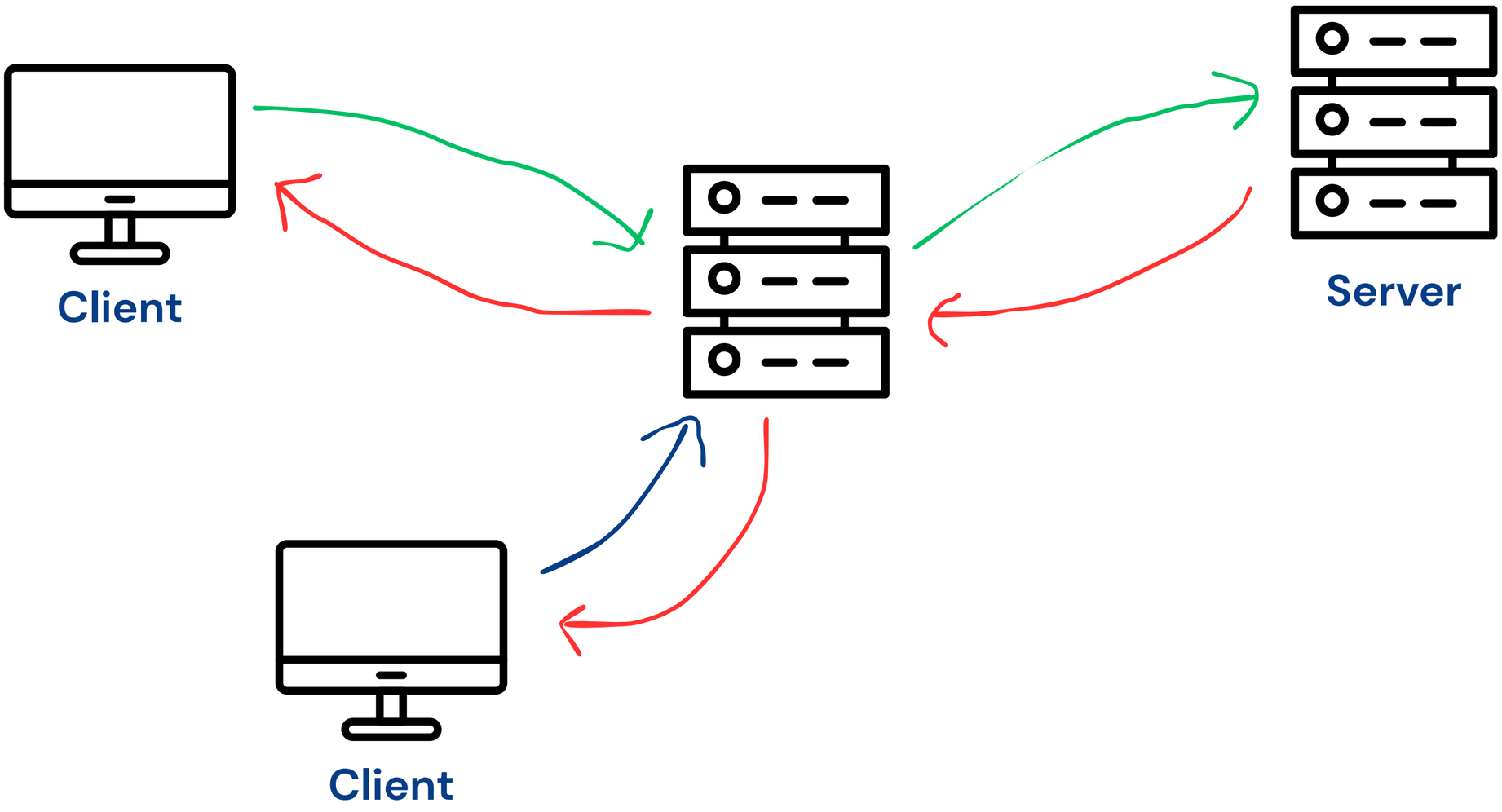
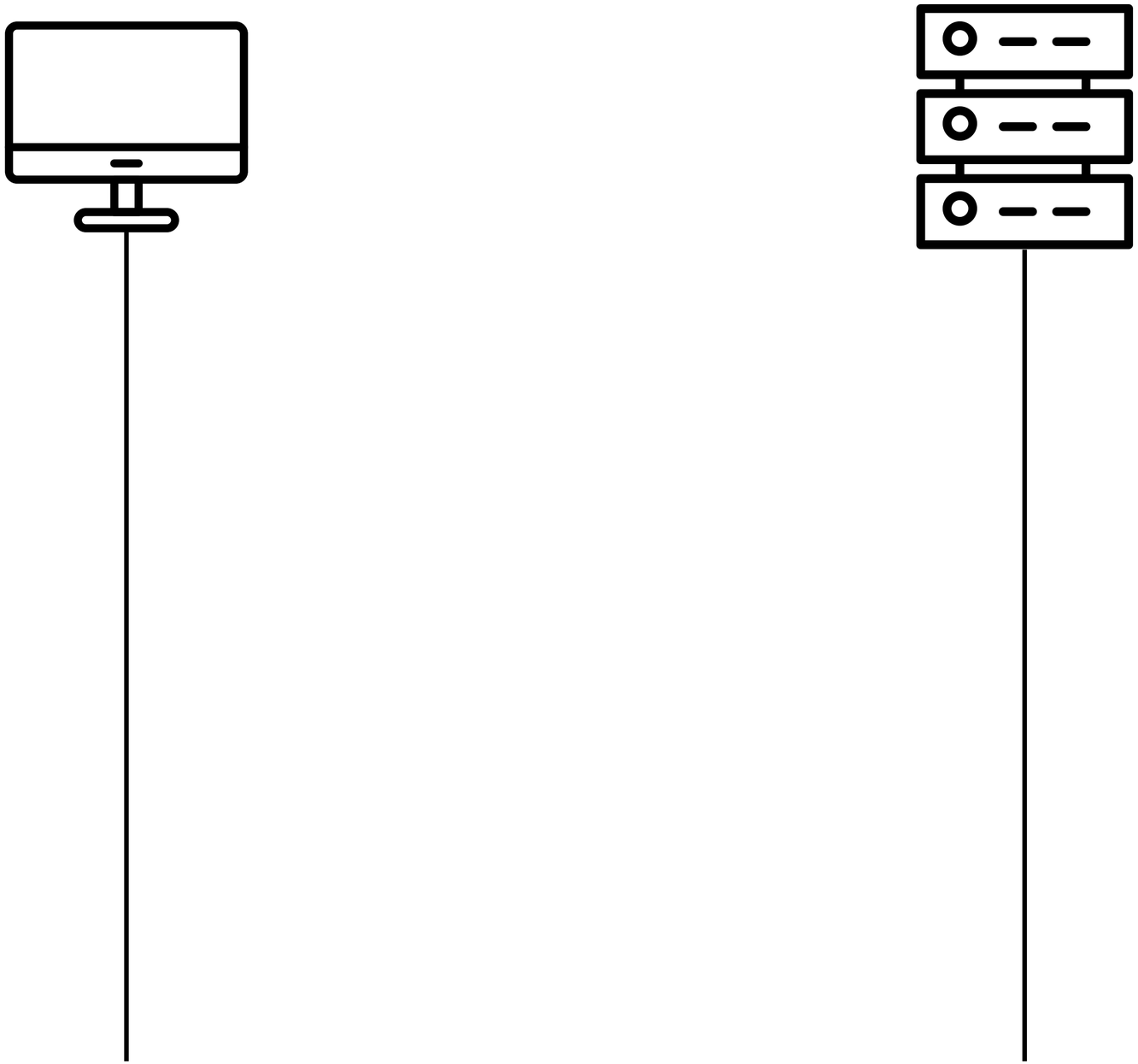


Increasing Performance (Browser Caching)

Suppose you have last accessed the webpage on <date> and your browser has cached it.



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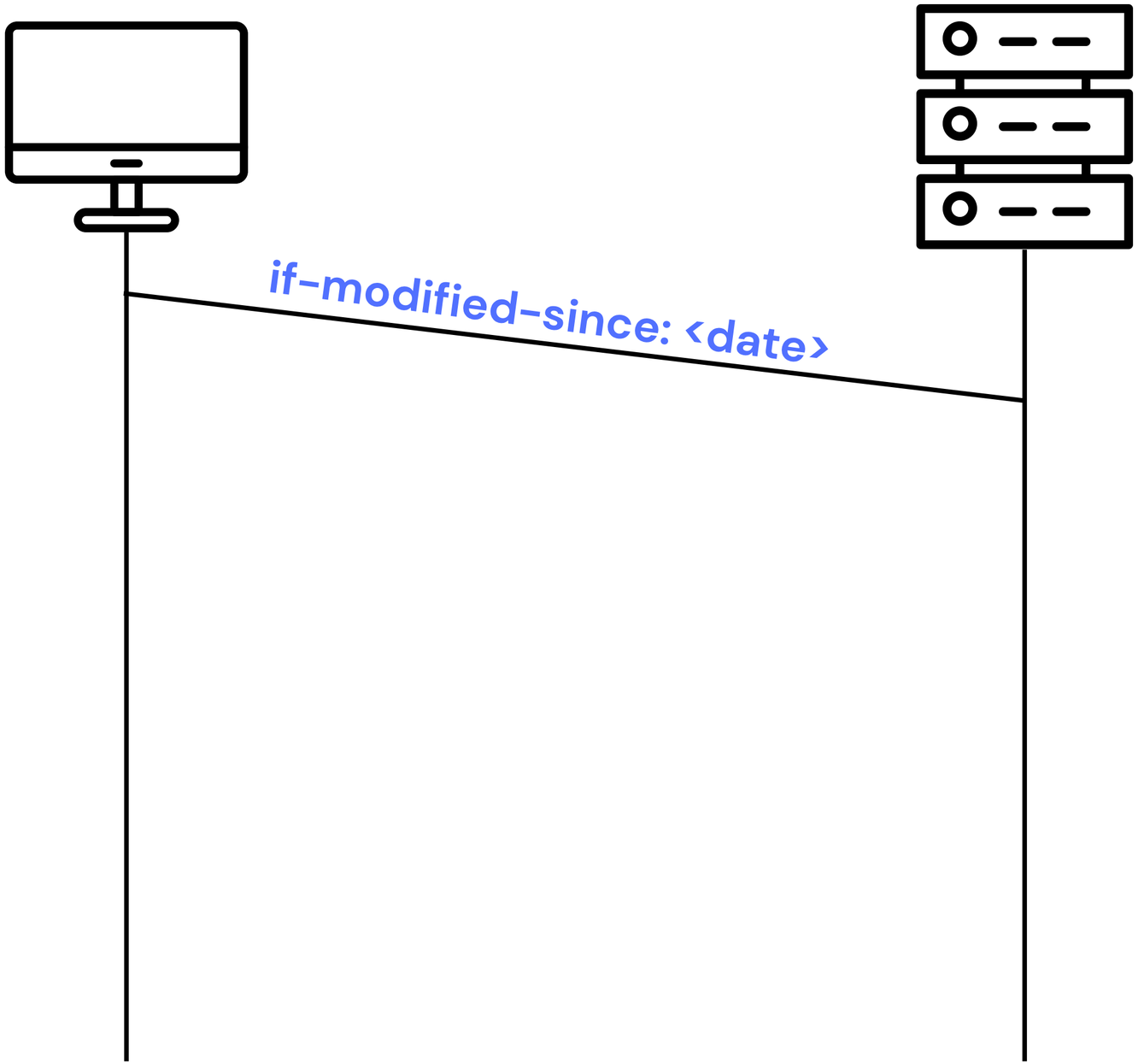

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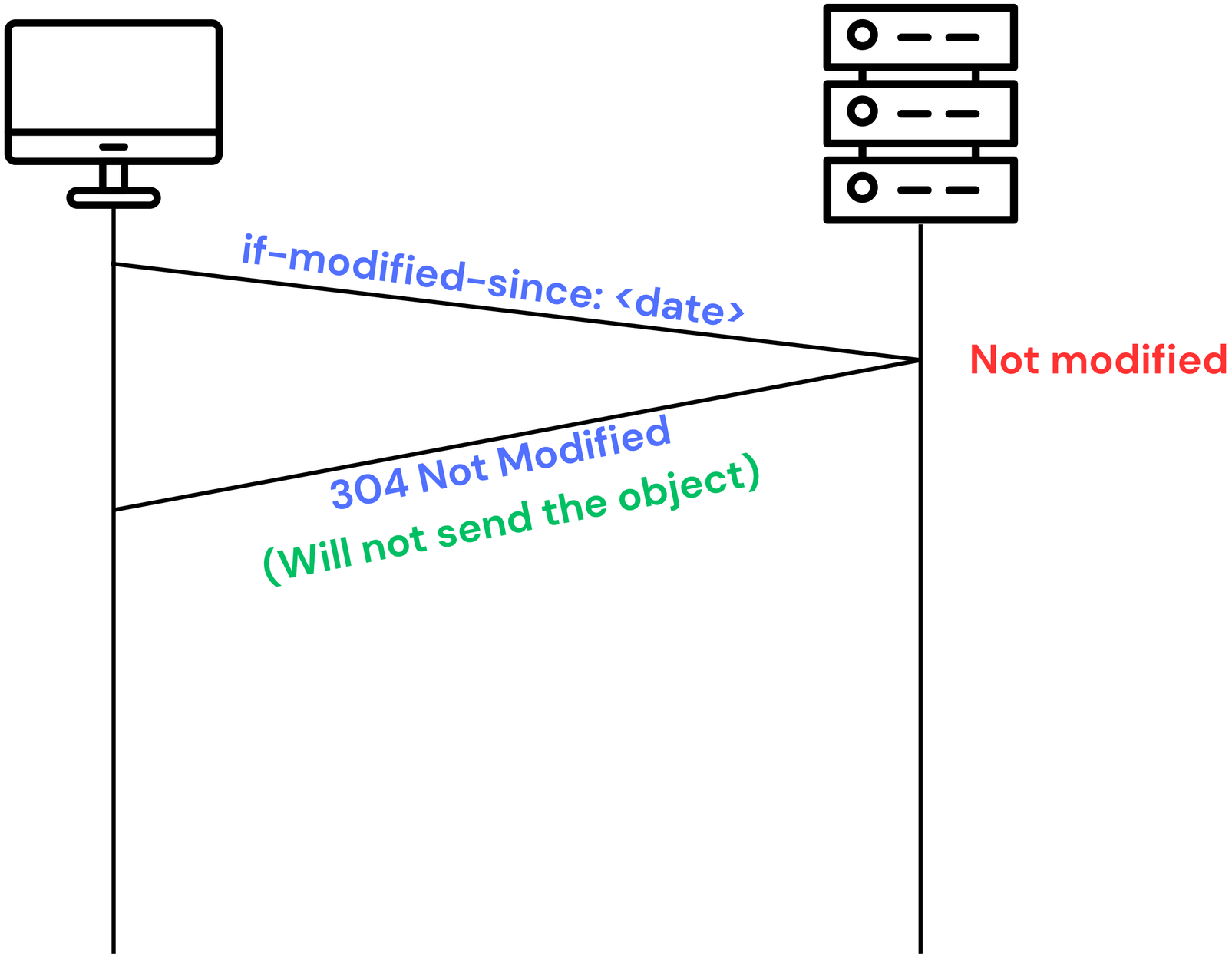

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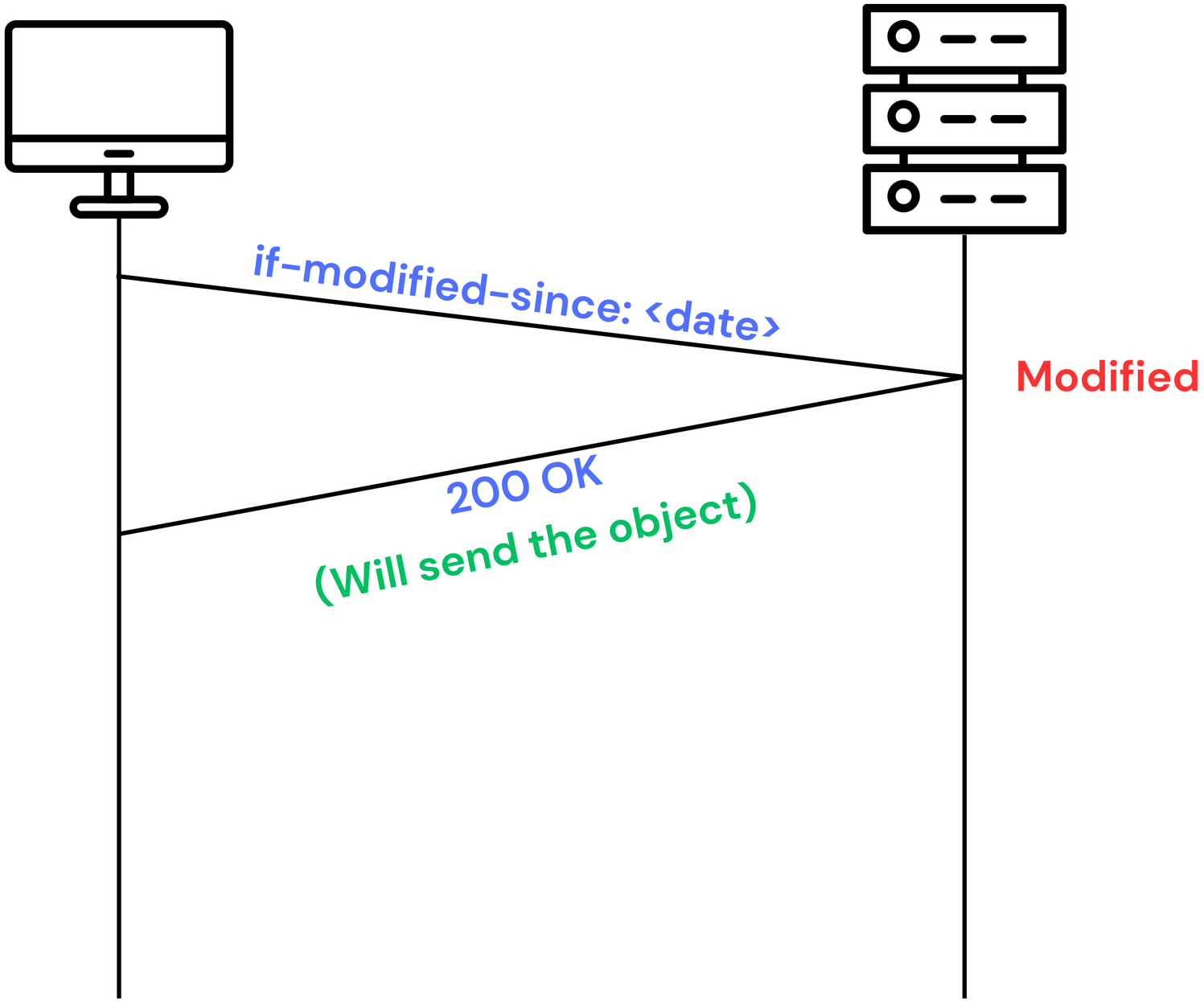

HTTPS

- Secured counterpart of HTTP.
- Uses TLS (Transport Layer Security)
- Provides encryption, authentication, and data integrity

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